

STREET LIGHT SYSTEM USING

ESP32

An energy efficient automated lightening architecture for smart cities presented by: *yogita paliwal*

From : GNIOT

Introduction:

Street lighting accounts for a major portion of municipal electricity consumption.

Traditional systems run on fixed timers or manual switches, leading to significant power waste.

Smart Street Lighting automates light control dynamically based on ambient sunlight and real-time movement.

ESP32 Microcontroller serves as the central brain, enabling edge control and IoT dashboard integration via built-in Wi-Fi.

Problem statement and objectives

Problem statement:

Continuous high-intensity lighting during late hours burns excess energy.

Manual tracking of faulty bulbs causes delayed maintenance and safety risks.

Objectives:

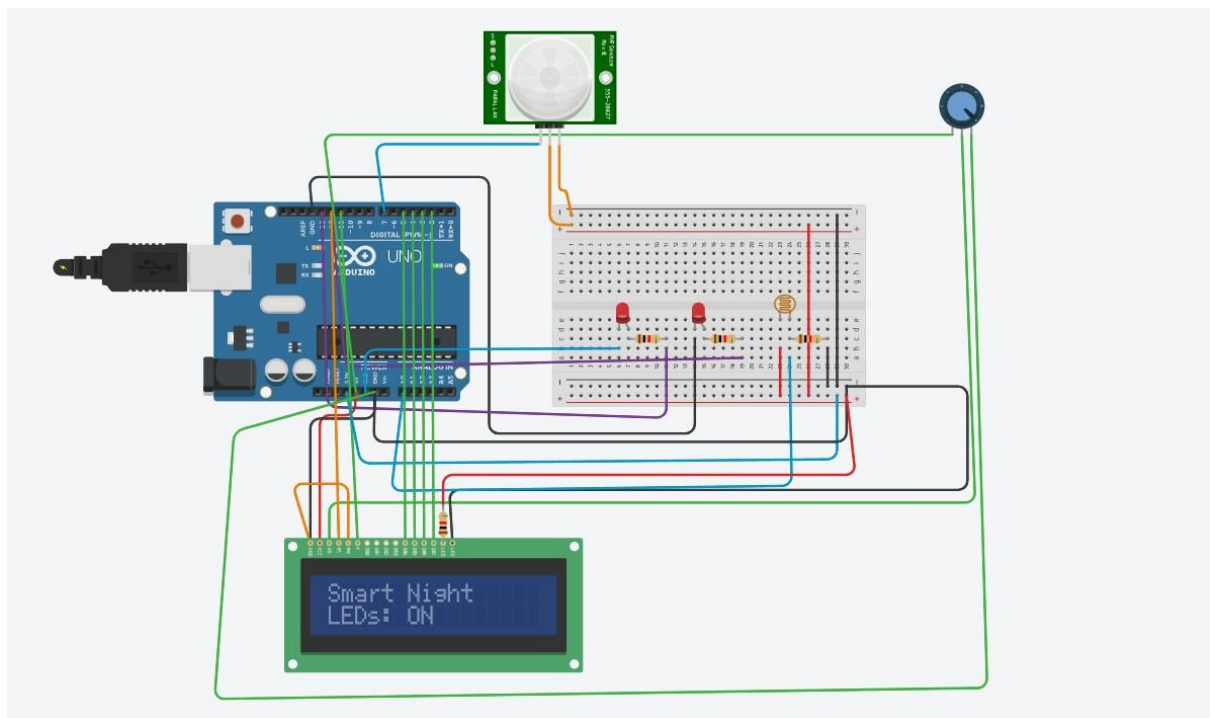
- **Energy Savings:** Automatically dim or turn off lights when no motion is detected.

Day/Night Sensing: Keep lights fully OFF during daylight hours using an LDR sensor.

- **Remote Monitoring:** Track live power metrics and light statuses over the cloud (Blynk / Adafruit IO / MQTT).

Automated Fault Detection: Send instant alerts when a light bulb fails.

System Architecture



- **LDR:** Measures ambient light intensity.

PIR/IR Sensors: Detect motion on roads and sidewalks.

- **ESP32:** Processes inputs, controls LED brightness via PWM, and uploads status data.

Key hardware components

Components and its functions:

- **ESP32 NodeMCU** Main microcontroller with dual-core processing and Wi-Fi/Bluetooth
- **LDR sensor** Detects day vs. night threshold
- **PIR / IR Sensors** Detects human or vehicle movement
- **LED Modules** High-efficiency street lamps with dimming support
- **Relay / MOSFET Module** Switches light power / handles PWM dimming

Power Supply 5V/12V DC power source

Working Methodology

- **Daytime Mode (High Ambient Light):**

LDR detects daylight ESP32 keeps street lights completely OFF.

Nighttime Standby Mode (Dark + No Motion):

LDR detects darkness ESP32 dims LEDs to low energy mode (~10%–20% brightness).

- **Motion Detection Mode (Dark + Motion):**

PIR/IR sensors detect approaching vehicle or pedestrian ESP32 boosts LED brightness to 100%.

Lights return to dimmed standby mode after a preset delay.

Cloud Reporting:

ESP32 streams state data and energy usage to the cloud dashboard via Wi-Fi.

Software & Cloud Integration

Development Environment: Arduino IDE .

Communication Protocols: MQTT / HTTP .

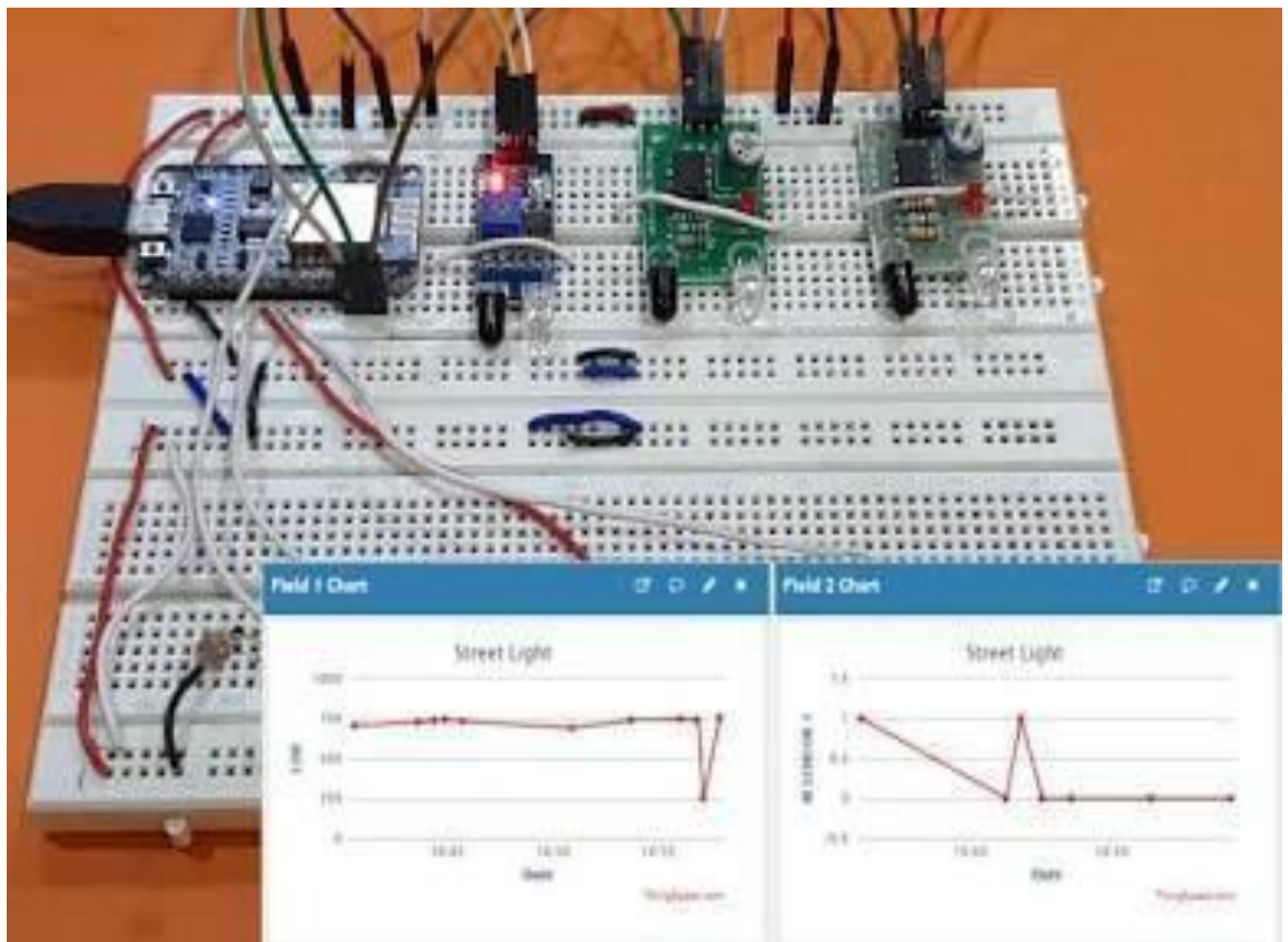
IoT Dashboard Features (Blynk / ThingsBoard):

Real-time toggles for Manual / Auto mode override.

Live brightness indicators and power consumption graphs.

Automated push notifications for fault detection.

MODEL IMAGE



Advantages & Impact

- **Energy Efficiency:** Reduces street lighting power consumption by up to 40%–60%.
- **Extended Bulb Lifespan:** Decreased runtime at 100% brightness extends LED life.
- **Low Operational Costs:** Automated fault alerts eliminate the need for manual night surveys.
- **Scalable Infrastructure:** Dual-core ESP32 easily integrates into larger smart-city IoT networks.

Conclusion & Future Scope

Conclusion

- The ESP32-based smart street lighting system offers a low-cost, automated solution for modern urban infrastructure.
- Combining motion dimming with IoT connectivity delivers significant energy and maintenance savings.
- **Future Enhancements**
- **Solar Integration:** Adding solar panels and battery monitoring for self-sustaining operation.
- **LoRaWAN Connectivity:** Extending range across vast rural and highway networks.
- **Air Quality Monitoring:** Adding environmental sensors (PM_{2.5}, CO₂) directly to light poles.